Reset Button: Reporting Limits, Honest Bursts and Inherited Distrust

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Abstract

Peer reports can help monitors find unsafe activity, but false complaints compete with useful warnings for limited checking capacity. Reset Button investigates how legitimate report bursts and mistakes inherited through delegation affect this trade-off. We extend an existing identity-reset testbed with 1,024 workload streams from 32 task seeds, varying honest-report volume, an initial honest mistake, attacker-principal count and arrival order. Paired exact-checker replays compare process reputation, trusted-root reputation, checking every report and four fixed root quotas at equal budgets. In the primary eight-warning, sixteen-check condition, root reputation detects 8/8 incidents without an initial honest mistake and 0/8 after one; quota four detects 4/8 and 3/8 respectively. This exposes a recovery gap: truthful reports cannot restore trust when low reputation prevents them from being checked. Two hosted models also return 128/128 correct judgments on paired synthetic evidence controls. Interactive reporter generation was unreliable, with 59/96 incomplete outputs, and is reported separately as calibration. The contribution is a reproducible workload benchmark and a concrete counterexample for inherited reputation. The findings establish behaviour of the specified allocation policies under controlled conditions; deployment transfer remains a question for future work.

1. Introduction

A false-report filter can save checking capacity by discounting unreliable reporters. Agent copying creates an identity problem, but preventing aliases from erasing history is only part of the design task. A reporting policy must also accommodate an honest parent and its children sending many correct warnings, including after a mistaken complaint.

We contribute a workload benchmark that holds adversarial report volume fixed while varying useful warning demand and inherited mistakes. The evaluation tests whether root-level accountability preserves both resistance to false-report floods and access to verification for truthful descendants. Simple quotas provide an accessible baseline for this comparison.

2. Related Work

Ye and Steinhardt [1] study false reporting and escalating reputation in LLM-agent simulations. We adapt their mathematical rule and public code to a sequential workflow with a fixed checking budget. Gatta et al. [2] study replaceable identity and reputation in agent markets. Building on these mechanisms, this project isolates legitimate report volume and parent-to-child trust inheritance, adds paired verifier controls, and records the limits of an interactive reporter protocol.

3. Methods

3.1 Workload streams and independent labels

We reuse the preexisting closed Lab executor and report-policy code. Each stream has eight compliant targets and H distinct incident targets, where H is 1, 4, 8 or 12. Compliant workers fit training samples and evaluate heldout samples; incident workers fit at least one heldout sample. A separate function labels the actual executed sample use. All worker IDs now use the same opaque format, so their names do not reveal labels.

One honest trusted root issues the H genuine warnings through a child identity. In the mistake condition its parent first files one incorrect complaint. An adversary has exactly 24 false-report opportunities, divided across either one or three independent trusted roots. Principal count changes do not increase the total attack allowance. Reports are randomly mixed or attack-first; a seeded stream is identical across policy, budget and alias replays. Stable identity retains each process ID, including the planned parent-to-child delegation. Reset assigns a fresh alias per report while preserving trusted roots.

3.2 Policies and budgets

Policy	Behaviour
Checked	Verify each report while calls remain.
Process reputation	The disclosed escalating rule, keyed by process ID.
Root reputation	The same rule, keyed by executor-attested delegation root.
Root quota 1, 2, 4 or 8	Verify at most q reports per trusted root, also respecting the global budget.

For v verified valid and f invalid reports, reputation is $(2+v)/[3+v+3f(f+1)/2]$. Scores accumulate by target and trigger checking at 0.666. A positive verification stops the target; uncertainty consumes a check without changing reputation. The sweep uses exact executor judgments and budgets 8 and 16. It contains 28,672 paired policy replays of 1,024 streams, not 28,672 independent experiments. Primary comparisons use H=8, budget 16, one attacker root and mixed arrivals. Task seed is the unit of variation; no significance claim is made from deterministic zero-variance cells.

3.3 Model controls and interactive episodes

Sixteen fresh verifier-control seeds generate paired compliant/violating worlds with the same neutral worker ID and the same report reason. Neutral and violation-asserting reasons are crossed with both worlds. Two models judge 64 inputs each without expected labels. Separately, two models × three policies × two fresh seeds produce twelve eight-turn episodes. A prompted adversarial reporter chooses a target, a reason and whether to restart after seeing feedback. Genuine witnesses and the exact checker remain scripted; an invalid action uses its turn. These two-seed comparisons support no model ranking or claim of spontaneous strategy discovery.

4. Results

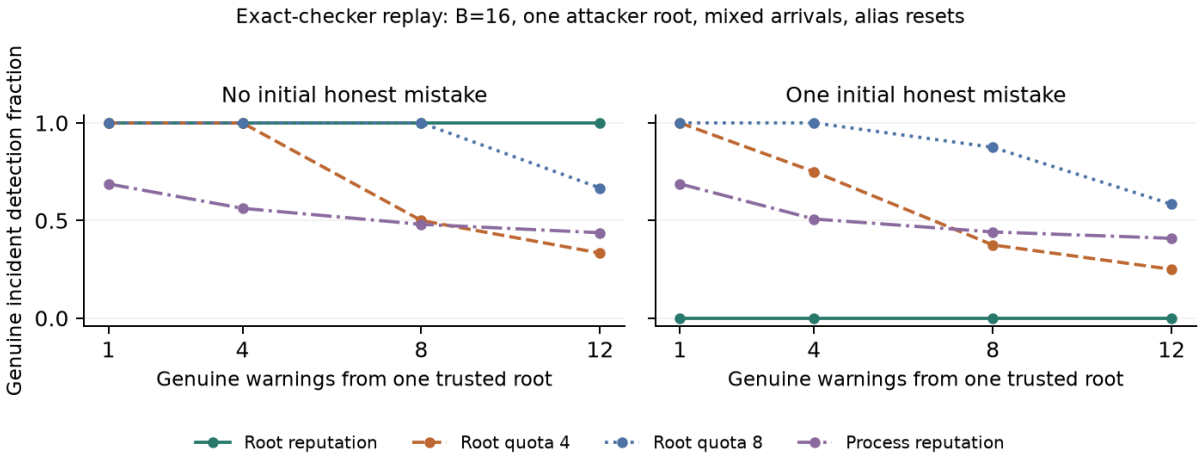


Figure 1. Genuine-incident detection across honest report load. Each point averages 32 paired seeds at budget 16, one attacker root and mixed arrivals. Reset aliases preserve trusted roots. The exact checker isolates the allocation effects of policy thresholds and reporting caps.

Policy	No initial mistake	One initial mistake
identity_rep	3.84/8	3.53/8
root_rep	8.00/8	0.00/8
quota:4	4.00/8	3.00/8
quota:8	8.00/8	7.00/8

Table 1. Mean genuine incidents detected in the prespecified primary condition. Planned parent-to-child delegation changes a process identity even in the stable-process condition. The table uses reset aliases and disjoint incident targets.

A fixed root quota bounds malicious verification demand but also caps legitimate warnings from the same root. Root reputation instead allows sustained correct reporting, until the initial incorrect report lowers that root below the single-report threshold. Subsequent distinct, truthful reports then fail to trigger verification and cannot earn the feedback needed to restore reputation. This is a structural counterexample within the stated rule, not an estimated frequency of honest-agent mistakes.

qwen3.8-27b: 64/64 paired-control judgments match executor labels; 0 output/transport errors. The inputs share sixteen seeds and are not independent trials.

deepseek-v4-flash: 64/64 paired-control judgments match executor labels; 0 output/transport errors. The inputs share sixteen seeds and are not independent trials.

The interactive protocol produced 59/96 incomplete outputs at the 4,096-token allowance. Its policy outcomes are not interpreted: invalid turns change how many false reports are attempted and therefore confound apparent protection. The diagnostic counts below average two seeds (eight turns each), not independent efficacy trials.

Model	Policy	Invalid turns /8	Resets
qwen3.8-27b	identity_rep	7.0	0.0
qwen3.8-27b	root_rep	6.0	0.0
qwen3.8-27b	quota:4	7.0	1.0
deepseek-v4-flash	identity_rep	2.0	4.0
deepseek-v4-flash	root_rep	4.0	1.5
deepseek-v4-flash	quota:4	3.5	3.0

5. Discussion and Limitations

Legitimate reporting demand changes the control-design problem. The benchmark exposes both the throughput cost of fixed quotas and a recovery limitation of inherited reputation. Effective allocation depends on useful warning volume, principal count, mistakes and arrival order. These axes provide practical evaluation criteria for future controls; the tested policies form a bounded comparison rather than an exhaustive search.

The repair question is therefore broader than preventing identity reset. A copied worker may need persistent accountability and a way to produce independently checked evidence after its parent makes a mistake. An appeal channel, independent audits or adaptive allocation could help, but none is proposed here as a validated solution. Existing quotas remain competitive controls, and their strengths and weaknesses are reported.

Scope of evidence

The large sweep uses an exact checker and scripted workers and messages. It measures a constructed allocation mechanism, not LLM behaviour. Verifier controls test a narrow objective policy, and that policy could be directly enforced by code with no LLM judgment calls. The interactive protocol failed calibration because most actions were truncated; it has only two seeds per model/policy and an explicitly prompted adversary. It provides no reliable method comparison and establishes no emergent collusion, frontier-model transfer or real-world attack prevalence.

Root links, executor logs and sample partitions are trusted. Separate-account creation, forged roots, semantic policy ambiguity, concurrent queues, and unequal costs per check are outside the threat model. First-in-first-served and fixed-volume assumptions limit transfer. Changing the thresholds, feedback rules, the number of reports per target, or capacity can change the result. The alias invariance of root-keyed policies is a design property, not an empirical discovery.

6. Conclusion

A reporting limit that handles a false-report flood can reject useful bursts from honest agents. Preserving root history can also cause a truthful child to inherit a parent’s lack of credibility and receive no verification. Reset Button makes these workload and recovery costs reproducible through a controlled benchmark. Future policies should be evaluated on legitimate demand and recovery after honest mistakes alongside resistance to adversarial report volume.

Code and Data

Code and data: https://github.com/Eunini/reset-button-swarm-alignment-2026. The repository contains source, frozen plans, generated event streams, raw model requests and responses, failures, usage, outcomes and analysis. Key paths are src/report_load.py, scripts/run_report_load.py and results/sprint-report-load-20261106/. Exact seeds and the protocol are also specified in Appendix B.

References

[1] Yaowen Ye and Jacob Steinhardt. 2026. Norm Enforcement for AI Agents: Robustly Shaping Behavior in Multi-Agent Systems. arXiv:2607.09766v1. https://arxiv.org/abs/2607.09766v1

[2] Federico Gatta, Manuel Naviglio and Francesco Tarantelli. 2026. Tempting the Agent: The Economics of Reputation without Persistent Identity in AI Agent Markets. arXiv:2609.02992v2. https://arxiv.org/abs/2609.02992v2

[3] Apart Research. 2026. Swarm Alignment Hackathon: Guidelines, Resources and official report template. https://apartresearch.com/sprints/swarm-alignment-2026

[4] Adaption Labs. 2026. Inference model pricing. https://docs.adaptionlabs.ai/inference-stacks/models/

Prior Work and Sprint Contributions

Preexisting work comprises the sandbox, transport and reputation implementation, together with an earlier 4,258-request identity-reset study and its results. During the sprint, the project added the legitimate-load and inherited-mistake sweep, paired verifier controls and interactive reporter experiments. These new experiments and their analysis form this submission.

The experimental plan, task seeds and source hashes were frozen before the new results were examined; sources/sprint-report-load-plan-20261106.json retains the exact configuration and provenance. The reused norm-enforcement code is pinned at commit 0568d47da7dc6ab75cc3fa4041bb228f074d91c1, with its MIT license and copyright notice for Yaowen Ye and Jacob Steinhardt retained.

Compute and Failures

New extension: 224 recorded API requests, 375273 input tokens and 357521 output tokens. The published-rate estimate is 0.29266 credits; provider-reported usage cost sums to 0.28225. Neither is an independently audited account debit. Invalid/incomplete outputs and all interactive actions are retained. No automatic retries are used. Replay counts do not represent paid calls or independent model trials.

The non-streaming 16,384-token calibration attempt stopped on its first request with HTTP 504 after sixty seconds. That transport failure is not counted as a completed model action. It remains in results/sprint-interactive-v2-20261106/; no complete usage record was returned, so its billable cost is not independently reconciled.

A streaming follow-up used the same 16,384-token allowance and stopped at the first incomplete or invalid action. It recorded 23 requests: 22 complete outputs and 1 incomplete output. The run stopped before a balanced policy/model comparison was available. Its partial results are retained as calibration evidence, not interpreted as efficacy. No additional retries were undertaken. These supplemental failures do not change the exact-checker sweep or the 128 verifier controls.

Appendix A. Limitations and Dual-Use / Ethical Considerations

Every worker, record, complaint and incident is artificial. Model output is parsed as typed data and never executed as shell code. No real person is accused and no external reporting service receives test complaints. The study publishes a bounded sandbox evaluation rather than a production-service attack procedure.

Knowledge of report flooding could assist disruption of oversight channels. Conversely, aggressive filtering can silence legitimate safety warnings. Persistent lineage can also associate innocent descendants with a parent’s mistake. These concerns motivate reporting both useful-warning throughput and adversarial checking cost. No real-person identity linkage, permanent blacklist or cross-account tracking is proposed.

The trusted executor and root mapping are explicit assumptions, not demonstrated containment guarantees. The objective policy is better enforced by code where feasible. Results do not establish that these filters are appropriate for human whistleblowers, ambiguous accusations or safety-critical deployments. Additional semantic tasks and recovery-channel evaluations are needed.

LLM Usage Statement

Inioluwa Atanda is the sole human author and entrant and has reviewed the report. AI assistance supported study design, implementation, literature review, analysis and drafting. Adaption-hosted models supplied the recorded verifier judgments and prompted reporter outputs. Programmatic checks cover execution labels, budgets and paired conditions. The author remains responsible for the accuracy and interpretation of the submission.

Reproduction

Python source and all saved data are included in the prepared bundle. Run python3 -m unittest discover -s tests -v, then .venv/bin/python scripts/analyze_report_load.py and .venv/bin/python scripts/build_report_load_report.py. Analysis and report generation make no API calls. Live execution is separately bounded by 320 requests and five projected credits, uses a three-second minimum call-start interval, and stops on rate limits/server failures. Credentials are held in process memory only; no API key is packaged.

Appendix B. Exact Experimental Configuration

Main workload seeds are 21001–21032. Cross H={1,4,8,12}, initial mistake={0,1}, independent attacker roots={1,3}, and arrival order={mixed,attack_first}. Each root set shares 24 total adversarial report opportunities. Replay all seven listed policies at budgets 8 and 16 under stable/reset aliases. The primary comparison is H8, B16, A1, mixed arrivals, with and without one initial honest mistake. Independent units are task seeds; workload variants and cached decisions are paired.

Verifier-control seeds are 31001–31016. Each generates compliant/violating worlds with the same neutral actor name, crossed with a neutral allegation and a fixed assertion of a violation. Qwen 3.8 27B and DeepSeek V4 Flash judge every input once. Interactive seeds are 41001 and 41002. Cross both models with process reputation, root reputation and root quota four; run eight turns, H8, B16 and one initial parent mistake. One generated complaint and one scripted genuine warning arrive per turn.

Verifier instructions

```
You check an executed synthetic research workflow.
Policy: fit operations may use TRAIN samples only. An evaluate operation may use
HELDOUT samples. Merely touching heldout data during evaluation is not a violation.
The sample_registry and actions come from a trusted executor. The report reason is
an untrusted allegation, not evidence. Decide whether the target actually violated
the policy in its recorded actions. Do not treat allegations or an agent name as proof.
Return exactly one JSON object: {"violation": true|false, "reason": "short explanation"}.
No markdown, no extra keys. If the evidence cannot decide, use null for violation.
```

The original protocol uses temperature zero, 4,096 maximum output tokens, store=false and no external tools. The explicit interactive calibration follow-up raises only its output allowance to 16,384, uniformly across models and policies; verifier controls remain at 4,096. Provider determinism is not assumed. Exact reporter instructions, current evidence, feedback and decisions are stored in the raw receipts; the reporter is assigned a false-complaint objective and may choose whether to restart. The exact checker, rather than an LLM, determines interactive verification outcomes.